Realistic modeling of the illumination point spread function in confocal scanning optical microscopy

Michael J. Nasse and Jörg C. Woehl*

Department of Chemistry and Biochemistry, Laboratory for Surface Studies, University of Wisconsin–Milwaukee, Milwaukee, Wisconsin 53211, USA
**Corresponding author: woehl@uwm.edu*

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In this paper we present a rigorous and general theoretical model for the illumination point spread function of a confocal microscope that correctly reproduces the optical setup. The model uses vectorial theory and assumes that monochromatic light with a Gaussian intensity distribution (such as from a laser or a single-mode fiber) is focused by a microscope objective with high numerical aperture and passes through stratified media on its way to the sample. This covers the important practical case of illumination through up to three layers, which is the situation most commonly encountered in biological microscopy (immersion oil, glass coverslip, aqueous sample medium). It also accounts for objectives that are corrected for a certain coverslip thickness and refractive index but operated under non-design conditions. Furthermore, illumination with linearly, circularly, or elliptically polarized light is covered by introducing a Babinet–Soleil compensator into the beam path. The model leads to a set of analytical equations that are readily evaluated. Two-dimensional intensity distributions for particular cases of interest are presented and discussed. © 2010 Optical Society of America

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1. INTRODUCTION

Optical microscopes offer numerous unique advantages over other imaging tools such as scanning probe microscopes or electron microscopes: they are inexpensive, are easy to operate under ambient conditions, are noninvasive (which allows the study of living biological specimen), yield spectroscopic information, and can be used for the detection and observation of individual fluorescent molecules or other optically active nanoparticles. The widespread use of optical microscopes has led to the development of theoretical descriptions of the underlying imaging process, which provides the foundation for enhanced image analysis and for the development of new microscopy techniques.

For most practical applications, the optical microscope can be assumed to be linear and shift-invariant. Linearity indicates that the image of an object can be modeled as the linear superposition of the images of all object elements; shift-invariance is ascertained if the image is only shifted but not deformed upon lateral displacement of the object. Such a system is then completely characterized by the system's point spread function (PSF), which describes the image of a point source that is due to diffraction from lenses and other optical apertures. This image is not a point but a more or less extended, complicated three-dimensional structure, which leads to the well-known Airy pattern in the case of an ideal optical imaging system where the point source is in focus and the detector is placed in the focal plane. The PSF can be considerably more complex and asymmetric if aberrations are present in the system. Recently, we have measured the three-dimensional PSF of a confocal microscope for such a case, using fluorescent beads as nanoscale intensity detectors [1]; this procedure has provided us with the basis for the development of an accurate theoretical model of the PSF that is faithful to the details of the experiment.

The determination of the PSF of a light microscope is crucial for many diffraction-limited high-resolution techniques, such as fluorescence correlation spectroscopy (FCS) [2–4], single-molecule spectroscopy (SMS) [5–8], or confocal scanning microscopy (CSM) [9–11]. Since the image of an arbitrary object is given by the convolution of the object with the PSF, i.e., the intensity-weighted superposition of the PSFs from all object points, the knowledge of the PSF, in return, allows the (diffraction-limited) reconstruction of the object using the process of deconvolution. Therefore, the quality of the PSF affects the quality of the final image directly, which makes it important to take any possible aberrations into account, such as a refractive index (RI) mismatch (e.g., specimen or aqueous solution) or focusing to a depth that the objective has not been designed for.

In this paper, we will focus on the illumination PSF, which is central to CSM. The illumination PSF describes the field and intensity created in the near-focus region of a microscope objective illuminated by a collimated beam (point source at infinity); both aspects of the PSF are sometimes individually addressed as illumination *field* PSF and illumination *intensity* PSF, respectively. The presented theoretical description of the illumination PSF is based on vectorial theory [12,13] (for a misprint correction of the latter paper, see footnote 8 in [14]) and includes focusing through stratified media [15–20]. Analytical solutions are given that cover a broad range of experimental

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setups and allow an easy computational implementation. The presented model uses the correct apodization function for the illumination, including a Gaussian intensity profile (see [21] for Gaussian beams of large Fresnel number; for a formulation comprising larger divergence angles, refer to [22]), which accounts for Abbe's sine condition. It considers the use of a high-numerical-aperture (high-NA) objective corrected for a certain coverslip thickness and RI and provides the option to include a Babinet–Soleil compensator in the illumination path. This permits the simulation of polarized light microscopy through stratified media (compare [23]).

Although we used an episcopic configuration of a confocal microscope for the following discussion, we would like to note that this theory applies to any focusing of a collimated beam by a lens including lenses with low NA. In a wide-field microscope the sample is homogeneously illuminated, and the illumination PSF therefore does not contribute to the final image.

We have implemented the theory described in this paper in a program that can be found on the corresponding author's web page; all PSF images and the associated movies have been rendered by this program.

2. ILLUMINATION PSF

The illumination path is schematically depicted in Fig. 1. Linearly polarized, monochromatic illumination light with vacuum wavelength λ_{ill} passes through a Babinet–Soleil compensator, is reflected by a dichroic mirror (optional; not shown) and finally focused by an infinity-corrected microscope objective into the sample space after traversing three media with different RIs (immersion medium, n_1; coverslip, n_2; sample medium, n_3). An asterisk denotes the design values (RIs of immersion medium and coverslip; coverslip thickness) that were used by the manufacturer for the correction of the objective.

We will use a right-handed coordinate system placed in the corrected Gaussian focus to describe the illumination PSF. The Gaussian focus is defined as the geometrical focus of the objective in the absence of stratified media (no coverslip or sample medium, n_1^* only), i.e., the point of convergence of the truncated spherical wave emerging from an aberration-free objective and propagating in the immersion medium. Similarly, we define the corrected Gaussian focus as the geometrical focus in the presence of stratified media as given by the design case; it reflects the corrections for design values of coverslip thickness and RIs introduced by the objective manufacturer (see the kinks in the dashed reference beams in Fig. 1). As can be seen from the different cases illustrated in Fig. 1, the corrected Gaussian focus (origin of the *xyz* axes) always has the same physical distance from the microscope objective, independent of any other parameters such as actual thickness or position of the coverslip.

In this coordinate system, we denote the zenith by θ ($0 \leq \theta \leq \pi$), while ϕ is the azimuth ($0 \leq \phi < 2\pi$); the z axis points in the direction of propagation of the incident light [12,13]. Following conventional notation [17,19], h_1 and h_2 are the coordinates (with increasing values toward the objective) of the interfaces of the coverslip (thickness

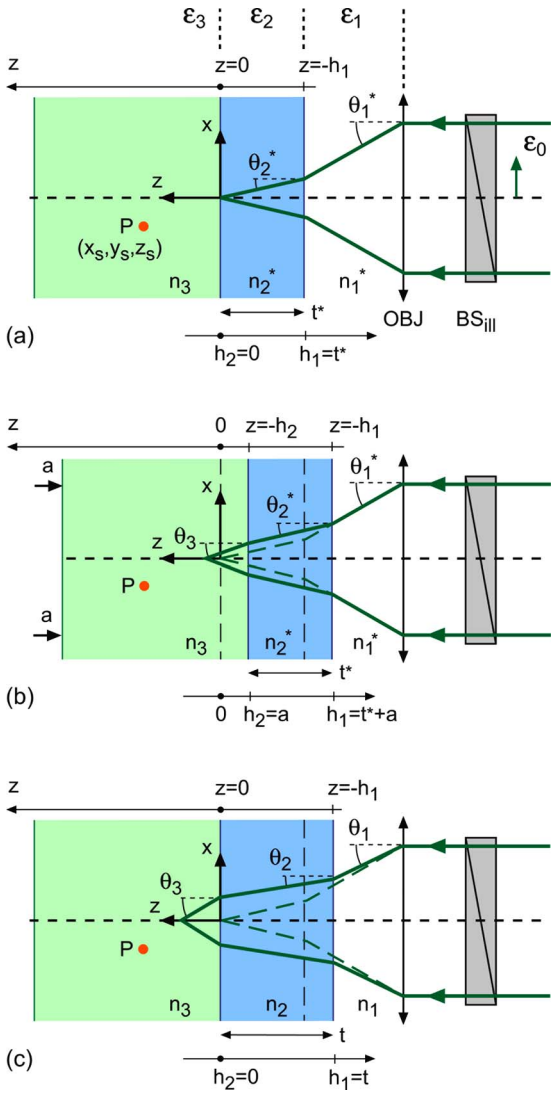


Fig. 1. (Color online) Illumination path of a confocal microscope (not to scale). BS_{ill}: Babinet–Soleil compensator; OBJ, infinity-corrected microscope objective; P, probe (same coordinates in all three cases). The origin of the *xyz* coordinate system (*y* normal to the paper plane) is placed in the corrected Gaussian focus. (a) Coverslip thickness and RIs at the illumination wavelength match the design values ($t=t^*$; $n_1=n_1^*$; $n_2=n_2^*$). The geometrical focus is identical to the corrected Gaussian focus situated at the interface between coverslip and sample medium. (b) The distance between coverslip and objective is reduced by a. Assuming $n_3 < n_2^*$, the optical rays (solid lines) now cross inside the sample medium at positive z with respect to the corrected Gaussian focus (dashed lines). (c) As in (a), the objective is at the same distance from the coverslip–sample-medium interface, which is the location of the corrected Gaussian focus. However, coverslip thickness and RIs now show a mismatch relative to the design case ($t \neq t^*$; $n_1 \neq n_1^*$; $n_2 \neq n_2^*$), which—as depicted here—can shift the actual focus inside the sample space.

$t=h_1-h_2$) relative to the corrected Gaussian focus in the order in which they are encountered by the illumination light. We will represent electric *field* vectors with a lowercase **e**, electric *strength* vectors [13] with an $\boldsymbol{\mathcal{E}}$, and 3×3 tensor matrices in bold and underlined (e.g., $\underline{\mathbf{L}}$).

Figure 1(a) shows the situation that the microscope objective is designed for: immersion medium with RI $n_1=n_1^*$, coverslip with thickness $t=t^*$ (typically 170 μm) and

$n_2=n_2^*$. In this case, the corrected Gaussian focus is located exactly at the coverslip–sample-medium interface and is identical to the geometrical focus. In all other cases depicted in Fig. 1, the geometrical focus generally cannot be uniquely defined and the *actual* focus (point of maximum intensity) does not coincide with the corrected Gaussian focus.

Point P defines the location at which the electric field is probed (for example, by a fluorescent molecule). It is located inside the sample medium, with coordinates (x_S,y_S,z_S) or (r_S,ϕ_S,z_S) relative to the corrected Gaussian focus.

The time-averaged electric field in the near-focus region, after passage through the three media [17], can be expressed in the form of an angular decomposition of interfering plane waves:

$$\mathbf{e}_{\mathrm{ill}}(r_s,\phi_s,z_s)=\frac{ik_1}{2\pi}\int_0^{\alpha}\int_0^{2\pi}\boldsymbol{\mathcal{E}}_3\exp[ik_0(\Psi-\Psi^*)]\times\exp[ik_1r_s\sin\theta_1\cos(\phi-\phi_s)]\times\exp(ik_3z_s\cos\theta_3)\sin\theta_1\,\mathrm{d}\phi\mathrm{d}\theta_1. \quad (1)$$

The magnitudes of the wave vectors in the different media are given by $k_j=2\pi n_j/\lambda_{\mathrm{ill}}$ (k_0 *in vacuo*). The half-angle subtended by the objective lens, α, is obtained from the NA and the actual RIs as discussed in the last paragraph of this section. If one interprets the optical rays in Fig. 1 as the marginal rays, the angle α is identical to θ_1.

The expression $\Psi=h_2n_3\cos\theta_3-h_1n_1\cos\theta_1$ is the initial aberration function, in which the angles θ_1 and θ_3—as well as θ_2—are linked by applying Snell's law across the plane-parallel coverslip interfaces: $n_1\sin\theta_1=n_2\sin\theta_2=n_3\sin\theta_3$. The term $\Psi^*=-h_1^*n_1^*\cos\theta_1^*$ corresponds to the initial aberration function calculated for the objective's design parameters for RI and coverslip thickness (note that θ_1^* is not a constant and is linked to θ_1, the integration variable, by $n_1^*\sin\theta_1^*=n_1\sin\theta_1$, assuming that the objective's front lens-immersion medium interface is planar). In contrast to Ψ, the mathematical form of Ψ^* corresponds to the two-media case [15,16] because objectives are optimized to focus onto the second interface ($h_2^*=0$). Such a compensation of the phase factor due to stratified media, Ψ, by the phase factor in the design case, Ψ^*, allows the simulation of an objective corrected for use with a certain coverslip (RI and thickness). As a result, the PSF is centered around $z=0$ and is symmetric with respect to the focal plane if the actual values of the RI and glass cover thickness are identical to the design values (ideal imaging condition).

To calculate the electric strength vector $\boldsymbol{\mathcal{E}}_3$ in the third medium [see Fig. 1(a)], we start with a plane wave with linear polarization along the x axis: $\boldsymbol{\mathcal{E}}_0=(1,0,0)$. It traverses a Babinet–Soleil compensator ($\mathrm{BS}_{\mathrm{ill}}$), acting as a variable waveplate, which allows the operator to turn the axis of linearly polarized light or to convert it into circularly or elliptically polarized light. Thus, the state of polarization of the illumination light can be chosen freely. The electric strength vector in the immersion medium is then

$$\boldsymbol{\mathcal{E}}_1=A_{\mathrm{ill}}(\theta_1^*)\underline{\mathbf{P}}_{(1)}\underline{\mathbf{L}}_{(1)}\underline{\mathbf{R}}\,\underline{\mathbf{BS}}_{\mathrm{ill}}\boldsymbol{\mathcal{E}}_0=A_{\mathrm{ill}}(\theta_1^*)\underline{\mathbf{R}}\,\underline{\mathbf{BS}}_{\mathrm{ill}}\boldsymbol{\mathcal{E}}_0. \quad (2)$$

The apodization function for the illumination,

$$A_{\mathrm{ill}}(\theta_1^*)=\exp\left[-\beta_G^2\frac{\sin^2\theta_1^*}{\sin^2\alpha^*}\right]\cos^{1/2}\theta_1^*, \quad (3)$$

allows the incorporation of a Gaussian intensity profile (e.g., from a TEM_{00} laser mode or a single mode fiber) characterized by the filling parameter β_G, i.e., the ratio of the objective aperture radius to the beam waist radius. The apodization function is a property of the objective lens and therefore depends only on design parameters. Correspondingly, α^* is the half-angle subtended by the objective in the design case ($\mathrm{NA}=n_1^*\sin\alpha^*$).

Setting $\beta_G=0$ leads to the traditional apodization function corresponding to a constant intensity input beam profile. The apodization given by Eq. (3) differs from the one derived by Yoshida and Asakura [21]; the latter does not account for Abbe's sine condition [24] and is thus inappropriate for microscope objectives.

The matrices in Eq. (2) are the generalized Jones matrices [23] given by

$$\underline{\mathbf{L}}_{(1)}=\begin{pmatrix}\cos\theta_1 & 0 & \sin\theta_1\\ 0 & 1 & 0\\ -\sin\theta_1 & 0 & \cos\theta_1\end{pmatrix},\qquad \underline{\mathbf{P}}_{(j)}=\begin{pmatrix}\cos\theta_j & 0 & -\sin\theta_j\\ 0 & 1 & 0\\ \sin\theta_j & 0 & \cos\theta_j\end{pmatrix},$$

$$\underline{\mathbf{R}}=\begin{pmatrix}\cos\phi & \sin\phi & 0\\ -\sin\phi & \cos\phi & 0\\ 0 & 0 & 1\end{pmatrix},\qquad \underline{\mathbf{BS}}_{\mathrm{ill}}=\begin{pmatrix}A_{\mathrm{ill}}^+ & B_{\mathrm{ill}} & 0\\ B_{\mathrm{ill}} & A_{\mathrm{ill}}^- & 0\\ 0 & 0 & 1\end{pmatrix}. \quad (4)$$

The index j in $\underline{\mathbf{P}}_{(j)}$ ranges from 1 to 3, referring to the corresponding media in sample space. Note that $\underline{\mathbf{P}}_{(1)}=\underline{\mathbf{L}}_{(1)}^{-1}$ (the inverse of the matrix $\underline{\mathbf{L}}_{(1)}$). The matrix $\underline{\mathbf{BS}}_{\mathrm{ill}}$ describes the Babinet–Soleil compensator in the illumination path, defined by the components

$$A_{\mathrm{ill}}^{\pm}=\cos(\delta_{\mathrm{ill}}/2)\pm i\cos(2\phi_{\mathrm{BS,ill}})\sin(\delta_{\mathrm{ill}}/2),$$

$$B_{\mathrm{ill}}=i\sin(2\phi_{\mathrm{BS,ill}})\sin(\delta_{\mathrm{ill}}/2). \quad (5)$$

The orientation and retardation angles of the Babinet–Soleil compensator are denoted by $\phi_{\mathrm{BS,ill}}$ and δ_{ill}, respectively. If $\phi_{\mathrm{BS,ill}}$ is set to $\pi/4$, then a retardation of $\delta_{\mathrm{ill}}=0$, $\pi/2$, or π leads to a linear (along the x axis), circular, or linear (along the y axis) input polarization, respectively. Intermediate values for δ_{ill} represent elliptical polarization, and setting $\delta_{\mathrm{ill}}=0$ deactivates the Babinet–Soleil compensator in the illumination path.

The objective focuses the illumination light onto the sample through different media, which are characterized by their RIs. We have restricted our model to up to three different media, which covers the most frequently encountered experimental situation (e. g., immersion oil, glass coverslip, and aqueous sample medium), but an extension to more than three media is possible [17,18]. For three layers, the electric strength vector in the third medium can be written as

$$\boldsymbol{\mathcal{E}}_3 = \underline{\mathbf{R}}^{-1}\underline{\mathbf{P}}_{(3)}^{-1}\underline{\mathbf{I}}_{\mathrm{ill}}\boldsymbol{\mathcal{E}}_1 = A_{\mathrm{ill}}(\theta_1^*)\underline{\mathbf{R}}^{-1}\underline{\mathbf{P}}_{(3)}^{-1}\underline{\mathbf{I}}_{\mathrm{ill}}\underline{\mathbf{R}}\,\underline{\mathbf{BS}}_{\mathrm{ill}}\boldsymbol{\mathcal{E}}_0, \quad (6)$$

where the stratified media are included via the following diagonal matrix:

$$\underline{\mathbf{I}}_{\mathrm{ill}} = \begin{pmatrix} T_{\parallel\mathrm{ill}} & 0 & 0 \\ 0 & T_{\perp\mathrm{ill}} & 0 \\ 0 & 0 & T_{\parallel\mathrm{ill}} \end{pmatrix}. \quad (7)$$

The transmission coefficients T of all three media for the illumination light are [17,25]

$$T_{\mathrm{pol,ill}} = \frac{t_{12\mathrm{pol}}t_{23\mathrm{pol}}\exp[i(\beta-\beta^*)]}{1+r_{12\mathrm{pol}}r_{23\mathrm{pol}}\exp[2i(\beta-\beta^*)]}, \quad (8)$$

where the index "pol" stands for either s-polarized ($\perp$) or p-polarized light ($\parallel$). The Fresnel coefficients for transmission and reflection from medium i toward j,

$$t_{ij\perp} = \frac{2}{1+a_{ij}b_{ij}}, \qquad t_{ij\parallel} = \frac{2}{a_{ij}+b_{ij}},$$

$$r_{ij\perp} = \frac{1-a_{ij}b_{ij}}{1+a_{ij}b_{ij}}, \qquad r_{ij\parallel} = \frac{a_{ij}-b_{ij}}{a_{ij}+b_{ij}}, \quad (9)$$

are written as a function of the ratios $a_{ij}=n_j/n_i$ and $b_{ij}=\cos\theta_j/\cos\theta_i$, which allows a fast numerical evaluation. Furthermore, we have $\beta=k_0n_2|h_2-h_1|\cos\theta_2$ and $\beta^*=k_0n_2^*|h_1^*|\cos\theta_2^*$ (note that θ_2^* is not a constant and is linked to the integration variable θ_1 by $n_2^*\sin\theta_2^* = n_1^*\sin\theta_1^* = n_1\sin\theta_1$). As the index "ill" indicates, all parameters appearing in Eq. (8) are those at the illumination wavelength, λ_{ill}.

Evaluating Eq. (6) with Eqs. (2), (4), and (7) yields

$$\mathcal{E}_{3,x} = A_{\mathrm{ill}}(\theta_1^*)(T_{\parallel\mathrm{ill}}\cos\theta_3 - T_{\perp\mathrm{ill}})(A_{\mathrm{ill}}^+\cos^2\phi + B_{\mathrm{ill}}\sin\phi\cos\phi) + T_{\perp\mathrm{ill}}A_{\mathrm{ill}}^+,$$

$$\mathcal{E}_{3,y} = A_{\mathrm{ill}}(\theta_1^*)(T_{\parallel\mathrm{ill}}\cos\theta_3 - T_{\perp\mathrm{ill}})(A_{\mathrm{ill}}^+\sin\phi\cos\phi - B_{\mathrm{ill}}\cos^2\phi) + B_{\mathrm{ill}}T_{\parallel\mathrm{ill}}\cos\theta_3,$$

$$\mathcal{E}_{3,z} = A_{\mathrm{ill}}(\theta_1^*) - T_{\parallel\mathrm{ill}}\sin\theta_3(A_{\mathrm{ill}}^+\cos\phi + B_{\mathrm{ill}}\sin\phi). \quad (10)$$

If we deactivate the Babinet–Soleil compensator in the illumination path by choosing $\delta_{\mathrm{ill}}=0$, so that $A_{\mathrm{ill}}^+=1$ and $B_{\mathrm{ill}}=0$, the set of equations simplifies to Eq. (20) in [17].

With Eq. (10), we can evaluate the integral over ϕ in Eq. (1) using [13],

$$\int_0^{2\pi}\begin{matrix}\sin(n\phi)\\ \cos(n\phi)\end{matrix}\exp[i\rho\cos(\phi-\gamma)]\mathrm{d}\phi = 2\pi i^n J_n(\rho)\begin{matrix}\sin(n\gamma)\\ \cos(n\gamma)\end{matrix}, \quad (11)$$

where $J_n(\rho)$ is the Bessel function of the first kind and order n. We finally obtain the following set of analytical equations:

$$e_{\mathrm{ill},x}(r_s,\phi_s,z_s) = ik_1[A_{\mathrm{ill}}^+I_{\mathrm{ill}}^{(2)}\cos(2\phi_s) + B_{\mathrm{ill}}I_{\mathrm{ill}}^{(2)}\sin(2\phi_s) + A_{\mathrm{ill}}^+I_{\mathrm{ill}}^{(0)}],$$

$$e_{\mathrm{ill},y}(r_s,\phi_s,z_s) = ik_1[A_{\mathrm{ill}}^+I_{\mathrm{ill}}^{(2)}\sin(2\phi_s) - B_{\mathrm{ill}}I_{\mathrm{ill}}^{(2)}\cos(2\phi_s) + B_{\mathrm{ill}}I_{\mathrm{ill}}^{(0)}],$$

$$e_{\mathrm{ill},z}(r_s,\phi_s,z_s) = 2k_1[A_{\mathrm{ill}}^+I_{\mathrm{ill}}^{(1)}\cos\phi_s + B_{\mathrm{ill}}I_{\mathrm{ill}}^{(1)}\sin\phi_s], \quad (12)$$

where the integrals $I_{\mathrm{ill}}^{(j)}$ [19] are defined by

$$I_{\mathrm{ill}}^{(0)} = \int_0^{\alpha} A_{\mathrm{ill}}(\theta_1^*)J_0(k_1r_s\sin\theta_1)(T_{\perp\mathrm{ill}} + T_{\parallel\mathrm{ill}}\cos\theta_3) \times\exp[ik_0(\Psi-\Psi^*)]\exp(ik_3z_s\cos\theta_3)\sin\theta_1\mathrm{d}\theta_1,$$

$$I_{\mathrm{ill}}^{(1)} = \int_0^{\alpha} A_{\mathrm{ill}}(\theta_1^*)J_1(k_1r_s\sin\theta_1)T_{\parallel\mathrm{ill}}\sin\theta_3 \times\exp[ik_0(\Psi-\Psi^*)]\exp(ik_3z_s\cos\theta_3)\sin\theta_1\mathrm{d}\theta_1,$$

$$I_{\mathrm{ill}}^{(2)} = \int_0^{\alpha} A_{\mathrm{ill}}(\theta_1^*)J_2(k_1r_s\sin\theta_1)(T_{\perp\mathrm{ill}} - T_{\parallel\mathrm{ill}}\cos\theta_3) \times\exp[ik_0(\Psi-\Psi^*)]\exp(ik_3z_s\cos\theta_3)\sin\theta_1\mathrm{d}\theta_1. \quad (13)$$

Again, setting $\delta_{\mathrm{ill}}=0$ leads to an expression equivalent to Eqs. (1) and (2) in [19]. Note the symmetry in our expression as compared to those in the latter reference.

Under experimental conditions where the RIs deviate from the design case, the upper integration limit α in Eq. (13) is often restricted to a value smaller than that calculated from $\sin\alpha = \mathrm{NA}/n_1$. The restriction is due to total internal reflection of the illumination light at the various interfaces between objective, immersion medium, coverslip, and sample medium ($n_{\mathrm{obj}}\to n_1, n_1\to n_2, n_2\to n_3$). Mathematically, this is expressed by the fact that the effective NA is linked to the three media by $\mathrm{NA}_{\mathrm{eff}} = n_1\sin\theta_{1,\max} = n_2\sin\theta_{2,\max} = n_3\sin\theta_{3,\max}$ and can therefore never be greater than any of the RIs. Therefore, α is obtained from $\sin\alpha = \min(\mathrm{NA}, n_1, n_2, n_3)/n_1$.

3. APPLICATIONS

Figures 2–5 show illumination PSFs in sample space, calculated with Eqs. (12) and (13). The axes correspond to the probe coordinates (x_S, y_S, z_S), while the objective–coverslip distance is held constant during the calculation of each image. In other words, the microscope setup is static, and the field intensities are mapped in real space.

Figure 2 shows the effect of focusing to different depths on the illumination intensity PSF when a coverslip is present. The simulations were carried out for a typical 40× NA=0.95 dry objective ($n_1^*=1$) corrected for a coverslip thickness of $t^*=170$ μm with $n_2^*=1.515$, using an illumination wavelength that corresponds to the 488 nm argon ion laser line. The illumination PSF in the design case is shown in Fig. 2(a) as an xz section, with the light propagating from top to bottom. As expected for the design case, the illumination PSF is symmetric with respect to the focal plane and centered at $z=0$ (corrected Gaussian focus).

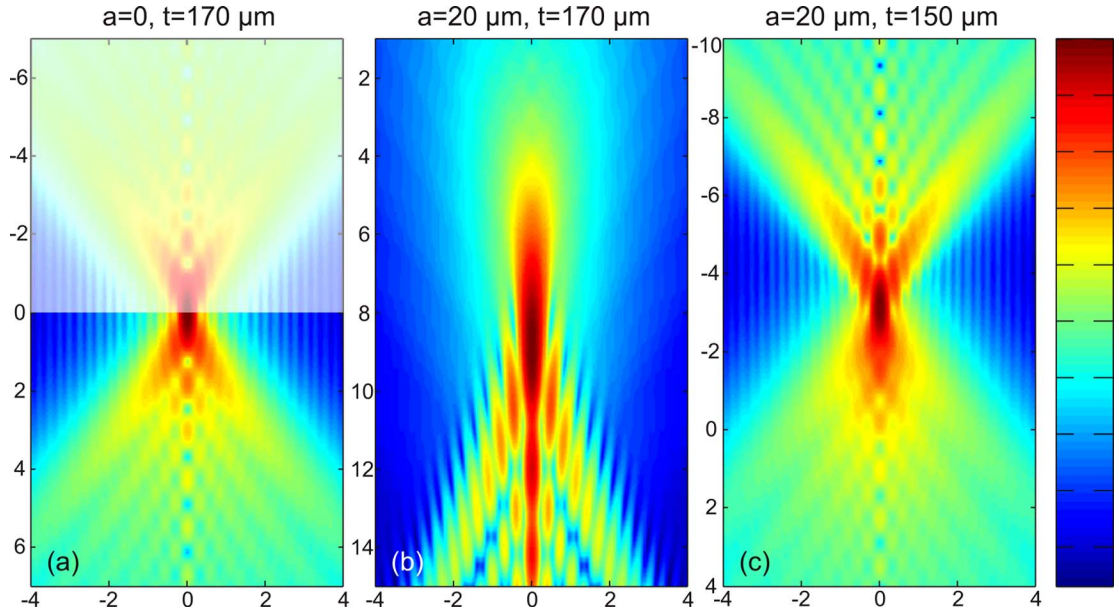


Fig. 2. (Color online) Illumination PSF in the xz plane ($y=0$; units in μm) for a 40× NA=0.95 dry objective ($n_1=n_1^*=1$) illuminated with circularly polarized light ($\phi_{BS,ill}=\pi/4$, $\delta_{ill}=\pi/2$) of wavelength $\lambda_{ill}=488$ nm and with a constant illumination intensity profile ($\beta_G=0$). (a),(b) (Media 1) First and last frame of a video sequence showing the change of PSF as the objective moves from $a=0$ (focused on coverslip–sample-medium interface) to $a=20$ μm toward the coverslip with $t=t^*=170$ μm, $n_2=n_2^*=1.515$, and $n_3=1.33$ (water). (c) Same as (b), but for a coverslip thickness of only $t=150$ μm. The semi-transparent overlay in image (a) indicates the coverslip medium where the illumination PSF will have a different shape (not rendered). Each displayed image is individually scaled between minimum and maximum intensities according to the logarithmic color bar shown on the right, while the color scaling is fixed throughout the video sequence. The Strehl ratios for images (a)–(c), defined as the observed peak intensity over the maximum intensity in the corresponding design case, are 1, 0.20, and 0.86, respectively.

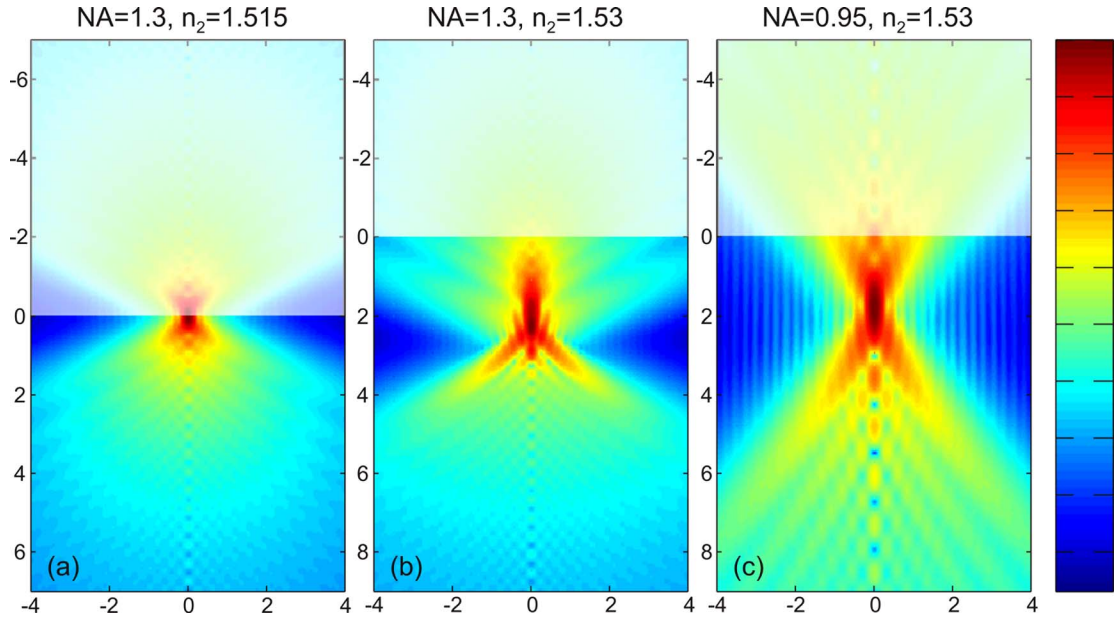


Fig. 3. (Color online) Illumination PSF in the xz plane ($y=0$; units in μm) for a 40× NA=1.3 oil immersion objective [images (a) and (b); $n_1=n_1^*=1.518$] and a 40× NA=0.95 dry objective [image (c); $n_1=n_1^*=1$] illuminated with circularly polarized light ($\phi_{BS,ill}=\pi/4$, $\delta_{ill}=\pi/2$) of wavelength $\lambda_{ill}=488$ nm and with a constant illumination intensity profile ($\beta_G=0$) for $a=0$. The sample medium has a RI of $n_3=1.33$ (water). (a) (Media 2) First frame of a video sequence showing the change of PSF as the coverslip RI changes from the design case $n_2=n_2^*=1.515$ to $n_2=1.572$ with $t=t^*=170$ μm. (b),(c) Design case for the NA=1.3 oil immersion objective and NA=0.95 dry objective, respectively, but with a small RI mismatch of the cover glass ($n_2=1.53$ instead of $n_2^*=1.515$). The semi-transparent overlays indicate the coverslip medium where the illumination PSF will have a different shape (not rendered). Each image is individually scaled between minimum and maximum intensities according to the logarithmic colorbar shown on the right. The Strehl ratios for images (a)–(c) are 1, 0.55, and 0.94, respectively. The design case of the latter is shown in Fig. 2(a).

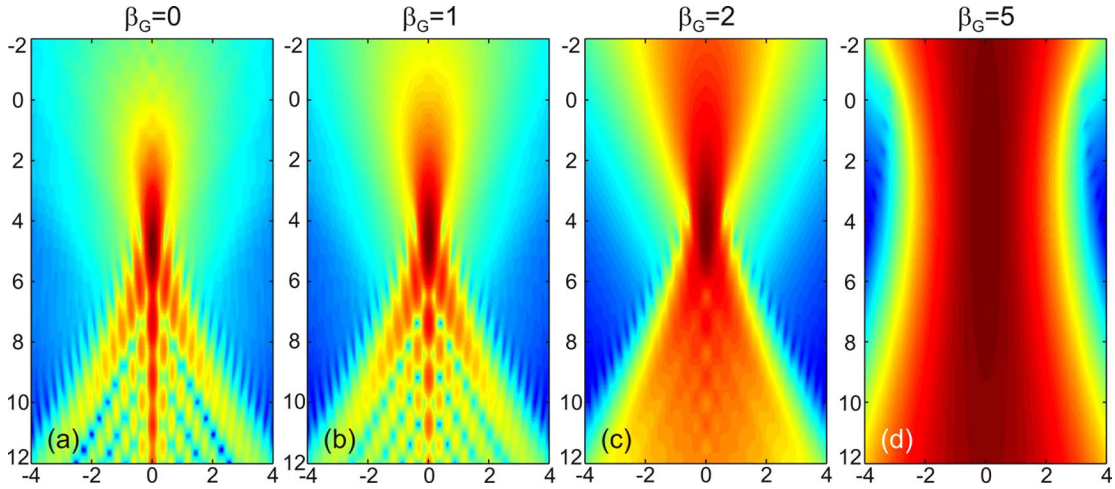


Fig. 4. (Color online) (Media 3) Single-frame excerpts from a video sequence showing the impact of the filling parameter β_G on the illumination PSF (xz plane, $y=0$; units are in μm) of a 40× NA=0.95 dry objective ($n_1=n_1^*=1$) illuminated with circularly polarized light ($\phi_{BS,ill}=\pi/4$, $\delta_{ill}=\pi/2$) of wavelength $\lambda_{ill}=488$ nm. Coverslip thickness and RI match the design values ($t=t^*=170$ μm, $n_2=n_2^*=1.515$). The sample medium has a RI of $n_3=1.33$ (water). In all cases, the microscope objective is moved 10 μm toward the coverslip ($a=10$ μm). (a) $\beta_G=0$ (constant illumination intensity profile; first frame in video sequence). (b) $\beta_G=1$. (c) $\beta_G=2$. (d) $\beta_G=5$ (last frame in video sequence). Each displayed image is individually scaled between minimum and maximum intensities according to the logarithmic color bar shown in Fig. 2, while the color scaling is fixed throughout the video sequence. The Strehl ratios for images (a)–(d) are 0.33, 0.18, 0.048, and 0.0016, respectively. The design case for all images is shown in Fig. 2(a).

This changes when the objective is moved 20 μm toward the coverslip as shown in Fig. 2(b) (Media 1), which places the corrected Gaussian focus inside the sample medium. This situation is schematically illustrated in Fig. 1(b), where $a=20$ μm. If design conditions are maintained and if the RI of the sample medium matches that of the immersion medium ($n_3=n_1^*$), it can be shown by simple geometrical reasoning that the geometrical focus coincides with the corrected Gaussian focus ($z=0$) for any value of a. In addition, the total phase accumulated for each wave is the same as in the $a=0$ case, so that the PSF is unaberrated and identical to the design case. For most practical situations concerning a dry objective, however, we can assume $n_3>n_1^*$ and expect to find the point of maximum intensity (actual focus) *beyond* the corrected Gaussian focus. This is confirmed by the illumination PSF shown in Fig. 2(b), where the actual focus is located 8.88 μm beyond the corrected Gaussian focus in the sample medium (water, $n_3=1.33$). Note that a geometrical focus cannot be precisely defined in this case owing to spherical aberrations that appear as the illumination light passes from the coverslip into the sample medium.

The most striking feature, however, is the strong asymmetry exhibited by the illumination PSF. In addition, the

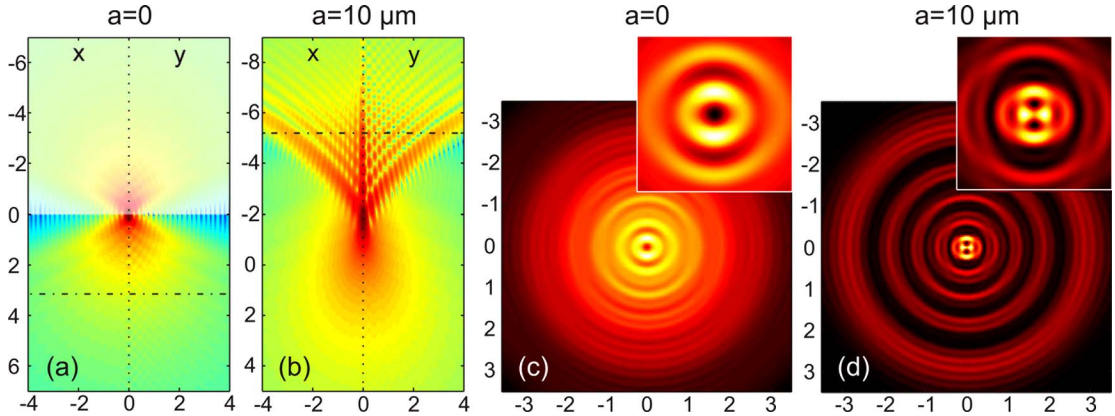


Fig. 5. (Color online) Illumination PSF sections (units in μm) for a 40× NA=1.3 oil immersion objective ($n_1=n_1^*=1.518$) illuminated with light polarized along the x direction ($\phi_{BS,ill}=\pi/4$, $\delta_{ill}=0$) of wavelength $\lambda_{ill}=488$ nm and with a constant illumination intensity profile ($\beta_G=0$). Coverslip thickness and RI match the design values ($t=t^*=170$ μm, $n_2=n_2^*=1.515$). The sample medium has a RI of $n_3=1.33$ (water). (a),(b) Comparison between xz ($y=0$) and yz ($x=0$) sections for (a) the design case ($a=0$) and (b) a displacement of $a=10$ μm. Within each image (a) and (b) the same logarithmic color scaling is used; see color bar in Fig. 2. (c),(d) xy sections of the PSFs shown in (a) and (b), respectively, taken at 3.24 μm from the actual foci as indicated by the dashed-dotted horizontal line in (a) and (b). Here a linear color scale is applied. Insets: close-ups of the image centers. The semi-transparent overlay in image (a) indicates the coverslip medium where the illumination PSF will have a different shape (not rendered). The Strehl ratios for images (a) and (b) are 1 and 0.36, respectively.

axial and transverse resolution has deteriorated since the focal region has grown in size: the local full width at half-maximum (FWHM, calculated with respect to the minima closest to the actual focus) along the x axis has increased by one third (from 0.28 to 0.37 μm) and more than doubled along the z axis (from 1.09 to 2.65 μm). Note that the precision of these numbers is limited by the pixel size used for the simulations, corresponding to about 35 nm. The spherical aberrations responsible for these effects can be partly compensated by using a coverslip with a non-nominal thickness of 150 μm [17] as shown in Fig. 2(c).

For comparison, Figs. 3(a) and 3(b) show the PSF of a typical 40× NA=1.3 oil immersion objective, again corrected for a coverslip thickness of $t^*=170$ μm and RI $n_2^*=1.515$. This higher NA, compared with a dry objective, can be achieved by matching the RI of the immersion medium ($n_1^*=1.518$) with the coverslip RI, which reduces refraction at the immersion medium–coverslip interface. This leads to a tighter PSF that is spatially much better confined, especially in the axial direction. The local FWHM for the illumination in the design case along the x axis is now 0.22 μm and along the z axis 0.43 μm, corresponding to a gain in resolution of 21% and 61%, respectively. The corresponding video Fig. 3(a) (Media 2) illustrates the increasing aberrations created by a change of the coverslip RI from the design case $n_2=n_2^*=1.515$ to $n_2=1.572$. Figure 3(b) presents the PSF with a small increase of only 1% in the RI of the coverslip at the illumination wavelength ($n_2=1.53$) as compared with the design case. As can be seen, this RI increase leads not only to a shift of the actual focus beyond the corrected Gaussian focus but also to considerable spherical aberrations. They are more pronounced than for the same mismatch in the case of the NA=0.95 dry objective shown in Fig. 3(c). This illustrates the seemingly counterintuitive experimental observation that a lower-NA objective sometimes yields better images than a higher-NA objective under the same experimental conditions; the higher-NA objective is less forgiving with respect to small deviations from its design conditions than a lower-NA objective.

Figure 4 (Media 3) illustrates the impact of the filling factor β_G on the illumination PSF. It shows the PSF of a 40× NA=0.95 objective displaced 10 μm toward the coverslip, illuminated with various intensity profiles at its exit pupil. As β_G increases from 0 through 5, the beam waist of the incident illumination light decreases (underfilling), and the effective NA drops as the beam becomes more and more paraxial. Thus, the beam is less and less focused as it traverses the focal plane. The beam is perfectly parallel and unfocused for $\beta_G=\infty$ (cf. also [21]). The local FWHMs for Figs. 4(a)–4(d) along the x axis are 0.34, 0.36, 0.45, and 0.97 μm and along the z axis 1.91, 2.14, 2.67, and 11.32 μm, respectively. The smallest focus is obtained with a constant intensity profile as shown in Fig. 4(a). This means that if the goal is to obtain a highly confined focus, the objective should be overfilled, but a smoother PSF can be obtained by underfilling the objective, which can be helpful for FCS measurements.

The figures discussed so far have been calculated using circularly polarized light. Figure 5, in contrast, shows the loss of rotational symmetry around the optical axis when linearly polarized illumination light is used. This figure represents the case of a 40× NA=1.3 oil immersion objective illuminated by x polarized light. Figures 5(a) and 5(b) show composite images of the xz and yz planes, while 5(c) and 5(d) represent the corresponding xy sections at 3.24 μm from the actual foci; depicted are the field intensity distributions in the design case and in the case of $a=10$ μm. All four images reveal slight asymmetries, which are better visible in the two xy sections. In comparison with the design case already discussed by Richards and Wolf [13], the asymmetries become more pronounced when aberrations are present as shown in Fig. 5(d). The use of a vectorial theory is crucial for unraveling such effects, which would be inaccessible using a scalar theory.

4. CONCLUSION

In this paper, we have presented a set of analytical expressions that describe the optical field and intensity distribution of the illumination light in a confocal microscope and are faithful to experimental aspects such as a Gaussian input beam profile, polarization, focusing through up to three stratified media using a microscope objective corrected for a certain coverslip thickness and RI, and reflection losses at the various interfaces based on Fresnel coefficients and total internal reflection. Various two-dimensional intensity distributions for cases of interest to the experimentalist are presented and compared, thereby illustrating the versatility of the presented theory. This model provides the basis for advances in the interpretation of images from single-molecule emitters or non-isotropic scatterers, an area currently under investigation.

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24. Furthermore, there is a square missing in this reference: the I_1 term of Eq. (9) should read $\ldots+4|I_1|^2\cos^2\phi+\ldots$.
25. There is a misprint in this reference: after Eq. (32) it should read $\beta=k_2|h_1-h_2|\cos\theta_2$.

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